



US 20250275699A1

(19) **United States**

(12) **Patent Application Publication**
TATE et al.

(10) **Pub. No.: US 2025/0275699 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **UNDERWATER EYE TRACKING SYSTEM
AND METHOD**

Publication Classification

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(51) **Int. Cl.**

A61B 5/16 (2006.01)
A61B 3/113 (2006.01)
A61B 3/14 (2006.01)
A61B 5/00 (2006.01)
B63C 11/12 (2006.01)
B63C 11/26 (2006.01)

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(52) **U.S. Cl.**

CPC **A61B 5/163** (2017.08); **A61B 3/113**
(2013.01); **A61B 3/14** (2013.01); **A61B 5/6803**
(2013.01); **B63C 11/26** (2013.01); **B63C 11/12**
(2013.01)

(21) Appl. No.: **18/804,495**

(22) Filed: **Aug. 14, 2024**

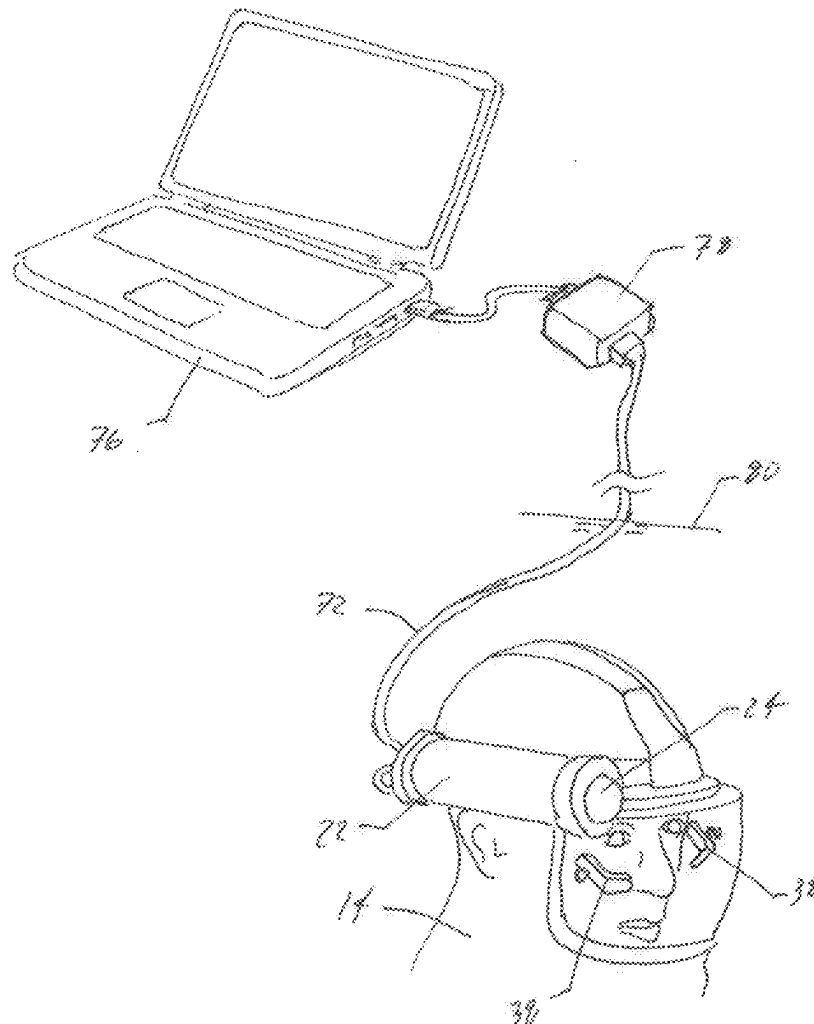
Related U.S. Application Data

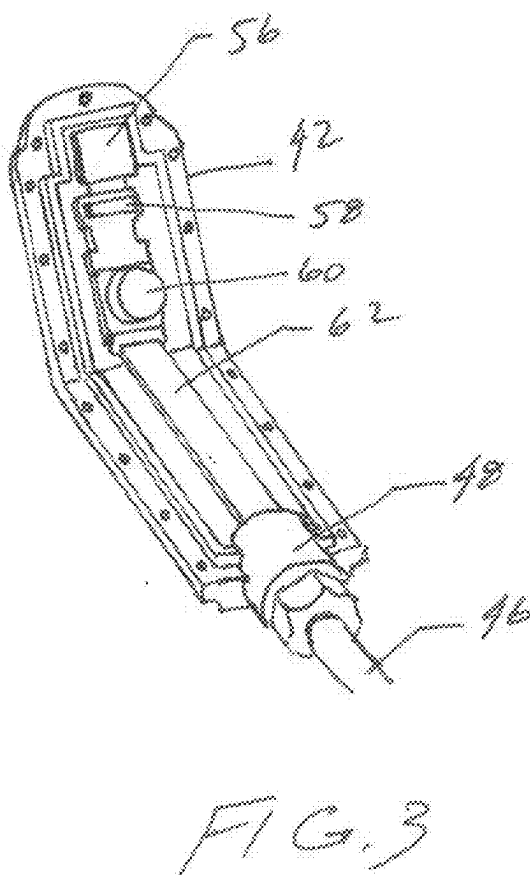
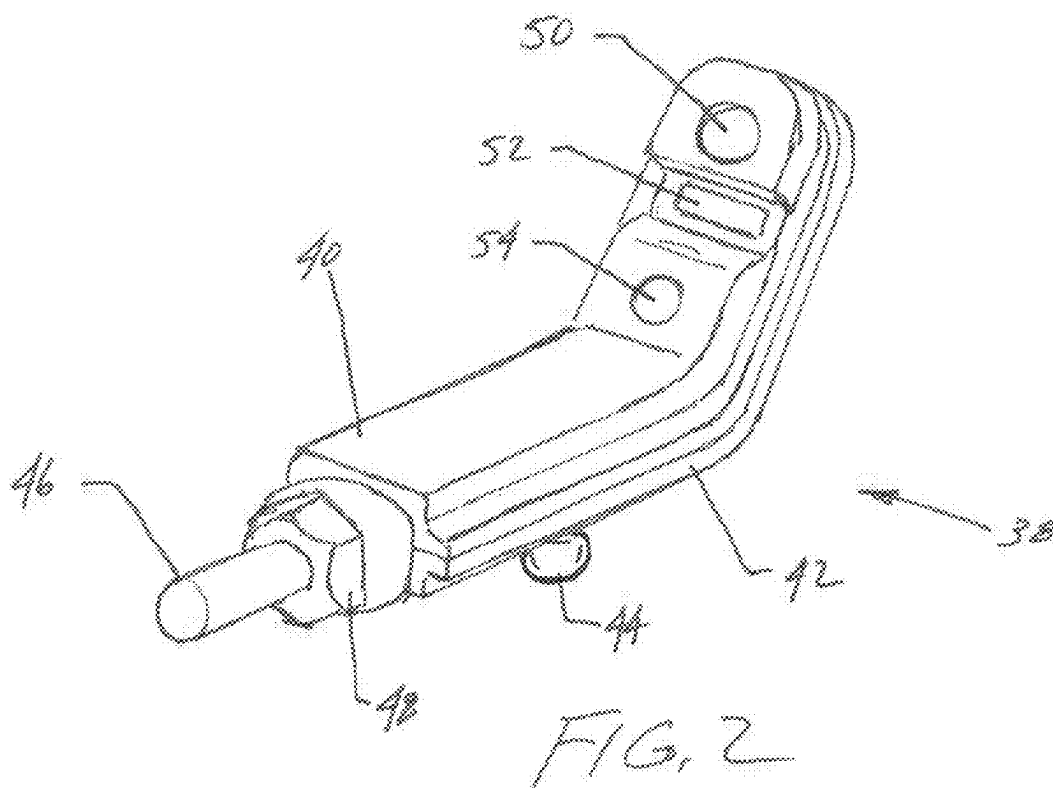
(60) Provisional application No. 63/532,492, filed on Aug.
14, 2023, provisional application No. 63/548,560,
filed on Feb. 1, 2024.

(57)

ABSTRACT

A system and method wherein a face mounted eye-tracking
system is mounted in a dive mask. The mask mounted
occulometric system has a novel configuration and enclosure
to provide in situ oculometrics and neurological assess-
ment.





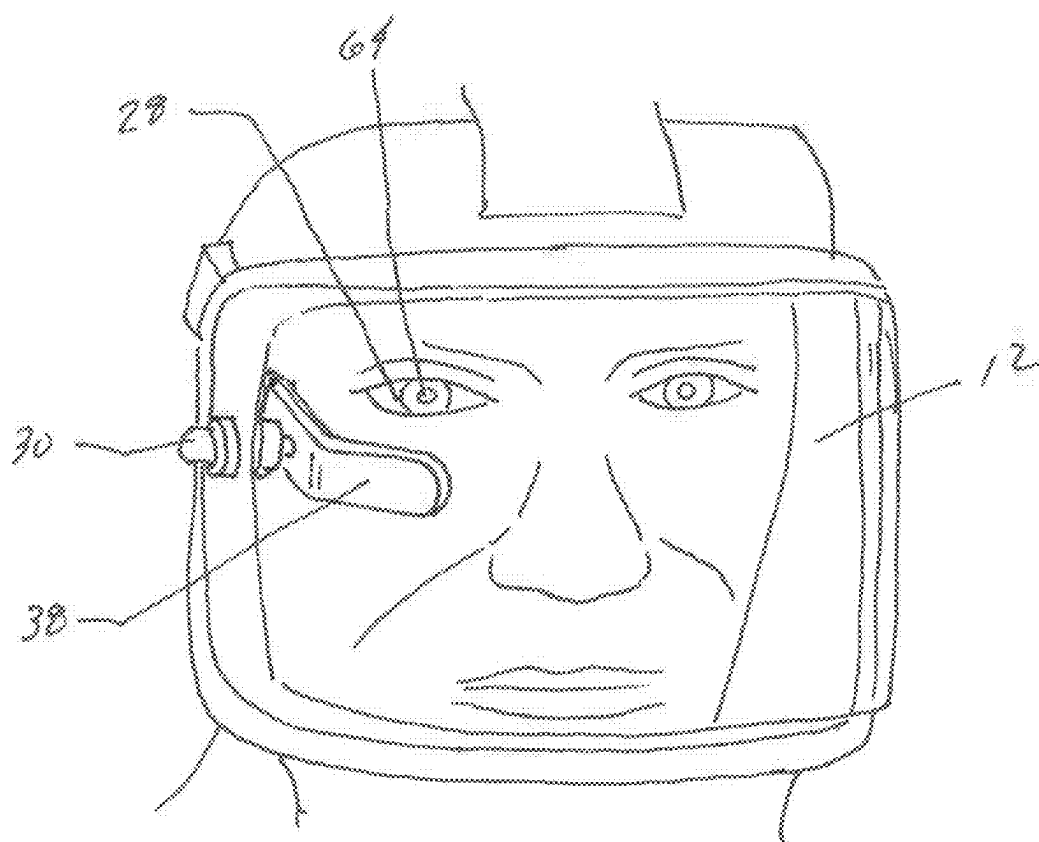


FIG. 4

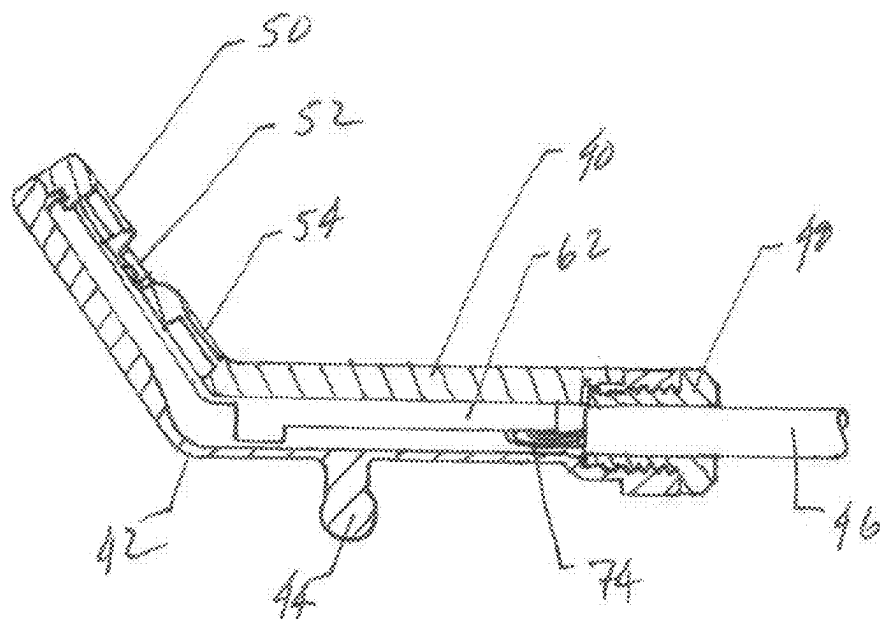


FIG. 5

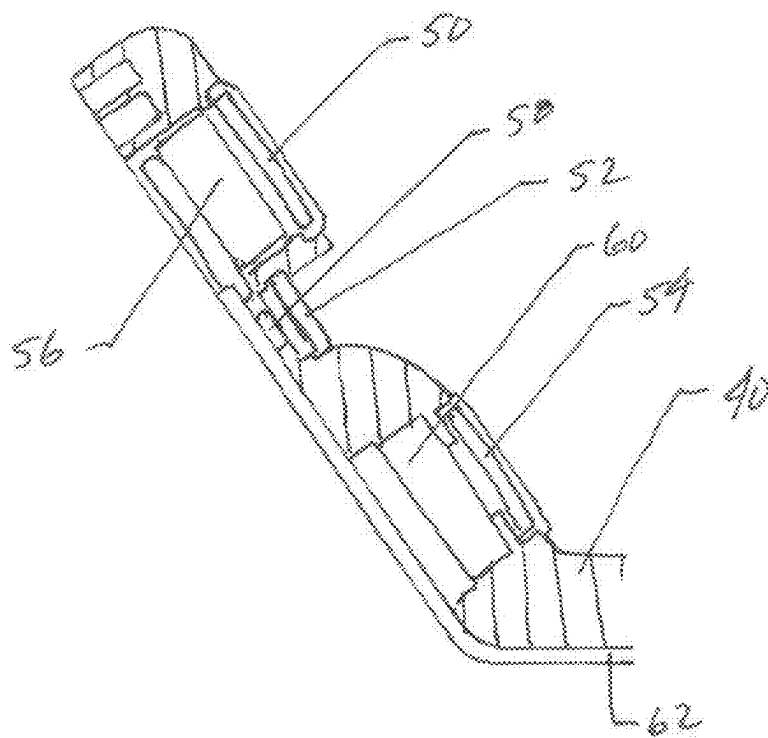


FIG. 6

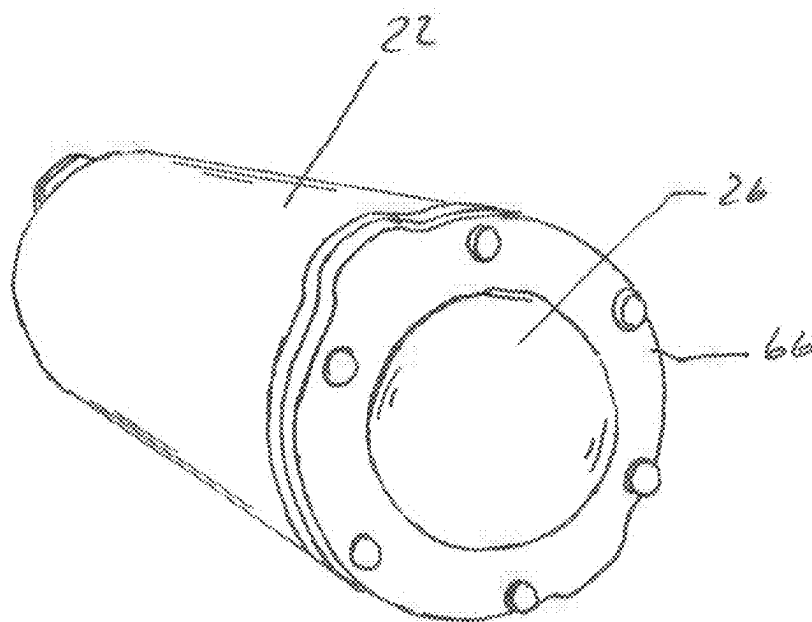


FIG. 7

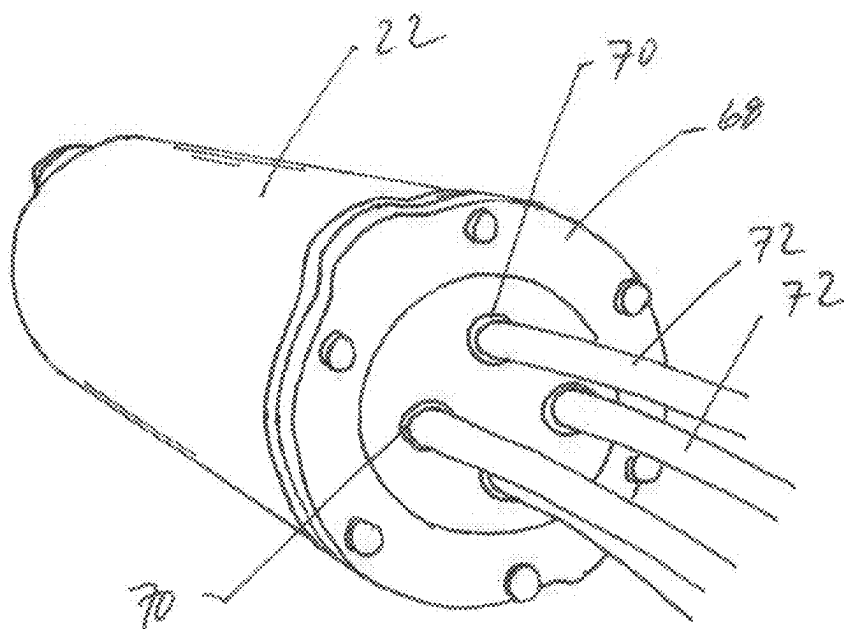


FIG. 2

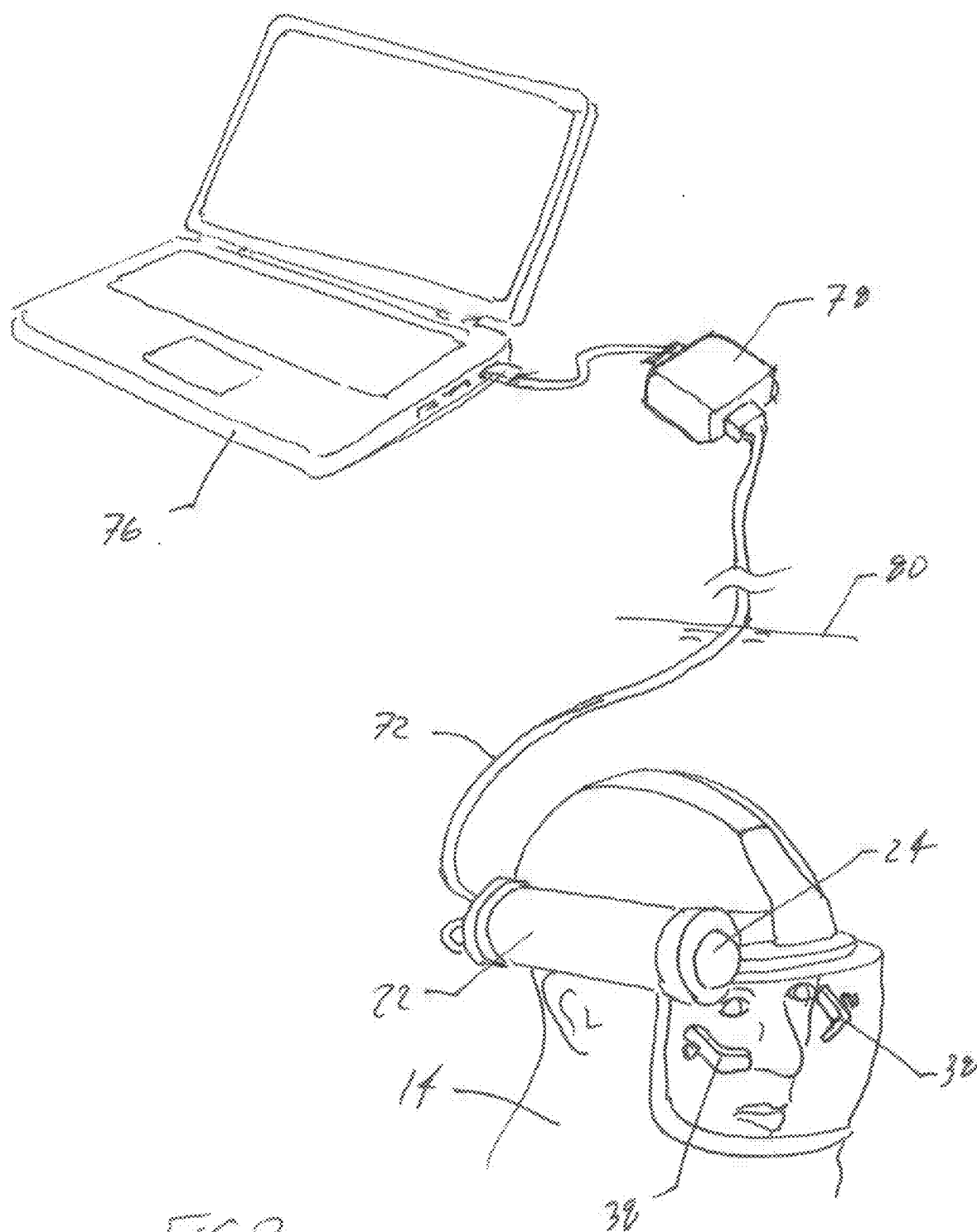


FIG. 9

UNDERWATER EYE TRACKING SYSTEM AND METHOD

CROSS-REFERENCES TO RELATED APPLICATIONS

[0001] This non-provisional patent application claims the benefit, pursuant to 37 C.F.R. section 1.53(c), of two previously filed provisional patent applications. The parent applications are: U.S. App. No. 63/532,492 filed on Aug. 14, 2023, and U.S. App. No. 63/548,560 filed on Feb. 1, 2024. Both provisional applications listed the same inventors as the present application.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Work on this project has been funded by the United States Government, Office of Naval Research, pursuant to Grantor No. N00014-20-1-2340.

MICROFICHE APPENDIX

[0003] Not Applicable

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0004] The present invention pertains to the field of human physiology monitoring. More specifically, the invention comprises a face mounted eye-tracking system that can provide in situ oculometrics and neurological assessment.

2. Description of the Related Art

[0005] The present invention provides the ability to monitor and assess a diver without unduly interfering with the diver's work. Divers are exposed to a range of environmental stressors that can negatively affect health status and outcomes. Environmental threats include nitrogen narcosis, oxygen toxicity, hypercapnia, and hypoxia. Such threats often arise from malfunctioning life-support equipment. Divers are trained to recognize the onset of many of these conditions, but the conditions themselves tend to impair cognitive functioning and self-recognition is therefore always risky (even for those conditions where self-recognition is possible).

[0006] Automated underwater operator health status monitoring is therefore an essential component of any mitigation strategy. Negative changes in health status must be identified as early as possible to extend the viable mitigation window prior to injury or incapacitation. The present invention provides a system and method for identifying these negative changes.

BRIEF SUMMARY OF THE INVENTION

[0007] The inventors have developed a face mounted eye-tracking system for use with a prior art dive mask-such as an INTERSPIRO DIVATOR MK11 dive mask. The mask-mounted oculometric system has a novel configuration and enclosure to provide in situ oculometrics and neurological assessment. Embedded LEDs provide spot checks of neurological status based on the pupil's response to light, focal stability, blink rate, and saccadic velocity. The system also possesses the ability to detect off-nominal eye states (such as closed eyes, irregular pupil location, or eye

movements). The system is adjustable, adapted to the diver and removable via the mask visor.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0008] FIG. 1 is a perspective view, showing a dive mask incorporating the invention.

[0009] FIG. 2 is a perspective view, showing an integrated tracking system according to the present invention.

[0010] FIG. 3 is a perspective view, showing the tracking system of FIG. 2 with the inner cover removed to show internal details.

[0011] FIG. 4 is a perspective view, showing a mask incorporating the integrated tracking system as shown in FIG. 2.

[0012] FIG. 5 is a sectional elevation view, showing internal details of the integrated tracking system.

[0013] FIG. 6 is a detailed sectional elevation view, showing more internal details of the integrated tracking system.

[0014] FIG. 7 is a perspective view, showing a first end of the electronics enclosure.

[0015] FIG. 8 is a perspective view, showing a second end of the electronics enclosure.

[0016] FIG. 9 is a perspective view, showing the connections used for remote monitoring of a diver using the inventive system.

REFERENCE NUMERALS USED IN THE DRAWING VIEWS

[0017]	10	dive mask
[0018]	12	mask window
[0019]	14	diver
[0020]	16	band
[0021]	18	superior strap
[0022]	20	diver
[0023]	22	electronics enclosure
[0024]	24	world view camera
[0025]	26	lens
[0026]	28	eye
[0027]	30	mount
[0028]	32	camera
[0029]	34	computer
[0030]	36	hub
[0031]	38	integrated tracking system
[0032]	40	inner cover
[0033]	42	outer cover
[0034]	44	ball mount
[0035]	46	cable
[0036]	48	waterproof via
[0037]	50	LED lens
[0038]	52	IR lens
[0039]	54	camera lens
[0040]	56	LED
[0041]	58	IR emitter
[0042]	60	camera
[0043]	62	flex circuit
[0044]	64	pupil
[0045]	66	first end cap
[0046]	68	second end cap
[0047]	70	waterproof via
[0048]	72	cable
[0049]	74	conductors
[0050]	76	laptop

[0051] 78 interface
 [0052] 80 water level

DETAILED DESCRIPTION OF THE INVENTION

[0053] The inventive device can be adapted for use in a wide variety of environments. This disclosure pertains to an embodiment configured for use in dive masks. FIG. 1 shows a version configured for use with the INTERSPIRO MKII dive mask, as marketed by Interspiro of Stockholm, Sweden. The inventive eye tracking device is integrated into the dive mask for use in submerged underwater field operations.

[0054] Dive mask 10 incorporates a large mask window 12. The mask window itself provides one or more mounts 30. These are often clamping devices placed through a bore in the window material. Band 16 passes around the head of diver 20. Superior strap 18 passes over the top of the head and attaches to the posterior portion of the band.

[0055] Electronics enclosure 22 is attached to the mask assembly. It may be mounted on the side of the band and/or attached to the frame of the mask window. World view camera 24 is contained within the electronics enclosure in this example. The world view camera is directed forward through lens 26. The world view camera preferably provides a wide field-of-view, so that a remote observer can see the environment directly in front of the diver's face. One or more forward-facing lights can be provided as well.

[0056] FIG. 1 includes a cutaway so the reader can visualize the interior of electronics enclosure 22. Monitoring and communication electronics are contained within enclosure 22. In the example shown, computer 34 and hub 36 are present. The computer performs local processing by retrieving and executing locally-stored software. Hub 36 facilitates communication with the eye tracking devices, the world view camera, external monitors, etc. Cables 72 pass into and out of the enclosure 22 through waterproof vias. The interior of enclosure 22 may be flooded with a suitable liquid to maintain neutral buoyancy (Air in the enclosure would tend to lift the mask assembly upward). Dielectric silicone liquid may be used to flood the interior.

[0057] It is preferable to provide a compact and integrated package for the emitters and sensors used in the inventive system. FIGS. 2 and 3 illustrate such a package. FIG. 2 shows integrated tracking system 38. Inner cover 40 and outer cover 42 are sealed together to encapsulate a set of enclosed components—typically a conventional LED, an infrared emitter, and a camera. The LED transmits light through LED lens 50. The IR emitter transmits through IR lens 52. The camera—which is typically an IR capable camera—views through camera lens 54. Cable 46 contains multiple electrically insulated conductors. These pass into the interior of integrated tracking system 38 through waterproof via 48.

[0058] FIG. 3 shows the assembly with inner cover 40 removed. The reader will note how LED 56, IR emitter 58, and camera 60 are mounted within the sealed interior. These components are electrically connected to cable 46 in this example by flex circuit 62. A sealing perimeter is provided to waterproof the assembly. Fasteners can be used to join inner cover 40 and outer cover 42. They may also be glued or otherwise fused together.

[0059] Waterproofing gaskets can be added to increase the pressure resistance. The internal components can optionally be potted into a water-resistant material.

[0060] FIG. 4 shows an exemplary mounting of integrated tracking system 38 inside mask window 12 of a diving mask. Mount 30 is preferably made adjustable to account for variations in facial anatomy. The tracking system 38 is positioned so that LED 56, IR emitter 58, and camera 60 are directed toward pupil 64 of eye 28.

[0061] FIGS. 5 and 6 show sectional views through the exemplary integrated tracking system 38. FIG. 5 shows how inner cover 40 and outer cover 42 clamp together to house the internal components. Conductors 74 emerge from cable 46 and connect to flex circuit 62. The flex circuit carries the signals to and from the internal components—the LED, the IR emitter, and the camera in this example. Waterproof via 48 seals the entrance of the cable.

[0062] Ball mount 44 extends laterally from outer cover 44. This ball mount can be clamped into an adjustable ball-and-socket mount that is attached to the mask window or some other mounting point on the mask assembly. The ball mount allows the integrated tracking system to be pivoted as desired and then locked into position.

[0063] FIG. 6 is a detailed view in the vicinity of LED 56, IR emitter 58, and camera 60. In this example, inner cover 40 includes recesses that capture LED lens 50 over the top of LED 56, IR lens 52 over the top of IR emitter 58, and camera lens 54 over the top of camera 60. These provide a waterproof seal for the operating depth range of the device.

[0064] FIGS. 7 and 8 show additional details of electronics enclosure 22. First end cap 66 mounts lens 26. This assembly is connected to the first end of the cylindrical electronics enclosure 22. FIG. 8 shows second end cap 68, which is attached to the second end of electronics enclosure 22. Multiple waterproof vias 70 are provided so that electrical cables 72 can pass through second end cap 68. Cables 72 connect the electronics inside the electronics enclosure 22 to one or more instances of integrated tracking system 38. They also typically connect the electronics to an external monitoring station (often on a surface vessel or platform).

[0065] The invention can be realized using a wide variety of hardware and software. An above-water laptop is typically used to acquire and process diver eye tracking data. This laptop runs PUPIL CORE open-source software (as marketed by Pupil Labs of Berlin, Germany) to acquire diver eye tracking data.

[0066] In the example shown in FIG. 1, computer 34 is a single board computer known as a RASPBERRY PI ZERO, marketed by Raspberry Pi, Ltd., of England. Hub 36 is a ZERO 4U four-port USB 3.0 hub as marketed by Adafruit Industries of New York, NY. This hub is used to integrate the various camera feeds and communicate to the laptop and the PUPIL CORE software. The hub is compatible with the RASPBERRY PI ZERO version 1.3. Other hubs may be substituted in other embodiments.

[0067] Camera 60 in the exemplary embodiment of FIG. 3 is a Model No. WS-NK001 S2.0infrared mini USB camera made in Guangdong, China. It features dual lenses. The IR capable camera is used for eye tracking. The world view camera is a 1080p wide angle camera. This is a low light camera for use in marine robotics. It is marketed by BLU-ROBOTICS of Torrance, California.

[0068] Dotstar and Neopixel are micro LEDs that can be used to elicit pupil latency measures. The LEDs are preferably controlled by the RASPBERRY PI ZERO computer. The LEDs can generate alternating flash patterns that are

useful in detecting pupil latency and measuring saccadic velocity (speed of movement during rapid involuntary eye movements).

[0069] The exemplary waterproof electronics enclosure shown in FIG. 1 is 50 mm×150 mm (2 inches×5.9 inches). It is available in either cast acrylic (rated to a depth of 250 m) or aluminum (rated to a depth of 950 m). The housing houses the USB hub for computer and sensor interfacing, the world view camera, and the RASPBERRY PI ZERO computer. The internal void space creates 2.58 Newtons of positive buoyancy, so it is desirable to offset the buoyancy force with added weight. For this reason, the aluminum version is preferred. It is also possible to fill the void with dielectric silicone oil to provide neutral buoyancy.

[0070] The acrylic dome mounted to one end of the enclosure is designed to be optically clear and suitable for use with the worldview low light USB camera. The four-port end cap shown in FIG. 8 closes the opposite end of the electronics enclosure. It has machined O-ring interfaces, a locking cord, anti-rotation features, and M10 threaded holes for the waterproof cable penetrators (rated to a depth of 950 m).

[0071] The materials used in the development of the mounts and enclosures are all low to no volatile organic compound emissions. This is important for the safety of the diver and the prevention of expansion and decompression stress on the equipment when alternating from submerged to unsubmerged operation. The SLS additive manufacturing process has been chosen for production because of the solid, layerless fabrication.

[0072] Optically clear visible and near-infrared light transmission lenses have been embedded in the enclosure to allow for the operation of imaging cameras and LED diodes.

[0073] FIG. 4 illustrates the integrated tracking system 38 mounted inside the dive mask and positioned to monitor the right eye 28. In many instances a second integrated tracking system will be mounted on the opposite side of the mask to simultaneously monitor the right eye (A dual installation is shown in FIG. 1).

[0074] FIG. 9 shows an operative environment in which the invention is used. Laptop 76 is used to monitor the status of diver 14. The laptop is typically maintained on a surface vessel or platform—well above water level 80. The laptop communicates with interface 78. Cable 72 connects interface 78 to the hub within electronics enclosure 22 mounted on the dive mask. The operator of the laptop can typically see the feed from world view camera 24, so the operator can see what is in front of the diver. Cables are used to connect the integrated tracking systems 38 to the hub within the electronics enclosure.

[0075] Integrated tracking systems 38 monitor the movements of the diver's eyes. The output of the cameras within the tracking systems can also be displayed to the laptop operator. The cameras 60 (see FIG. 3) are preferably capable of monitoring into the infrared range. IR light can be produced by IR emitters 58 as needed. LED 56 provides visual spectrum light for other actions—such as testing the reaction of the diver's eyes to light stimuli.

[0076] The use of the IR emitter and the IR-capable camera allows the diver's eye movements to be tracked without interfering with the diver's vision. A diver often works in a low-light environment and shining visible-spectrum light into the eye is undesirable. The use of IR light for

eye tracking is therefore advantageous because eye tracking can proceed without the diver even being aware of it.

[0077] The hardware can thus be used to monitor for threats to the diver's health status. As explained previously, divers are exposed to a range of environmental stressors that can negatively affect health status and outcomes. Environmental threats include nitrogen narcosis, oxygen toxicity, hypercapnia, and hypoxia. Such threats often arise from malfunctioning life-support equipment. Underwater operator health status monitoring is an essential component to any viable mitigation strategy. Negative changes in health status must be identified as early as possible to extend the viable mitigation window prior to injury or incapacitation.

[0078] The LEDs 56 provide spot checks of neurological status based on the pupil's response to light. The eye tracking camera can be used to measure focal stability, blink rate and saccadic velocity (velocity of involuntary eye movements). The system also possesses the ability to detect off-nominal eye states (i.e., closed eyes, irregular pupil location or unusual eye movements). The system is adjustable, adapted to the diver and removable via the mask visor.

[0079] For the installations employing two integrated tracking systems (one for each eye), the LEDs and camera can be used to evaluate pupillary reflexes for both eyes. The LEDs in each integrated tracking system 38 are configured to project light onto the proximal eye but not the distal eye. This allows the system to test for the pupillary light reflex. The system 38 proximate the right eye can project light into the right eye. The system 38 proximate the left eye can then monitor the left eye to see if the left pupil contracts in response to light being applied to the right eye. An abnormal response is useful in monitoring for neurological impairment of the diver.

[0080] The preceding descriptions contains significant detail regarding the novel aspects of embodiments of the present invention. It should not be construed, however, as limiting the scope of the invention but rather as providing illustrations of the preferred embodiments of the invention. Thus, the scope of the invention should be fixed by the claims ultimately presented, rather than by the examples given.

Having described our invention, we claim:

1. A method for monitoring the health status of a submerged diver wearing a mask, comprising:

- (a) providing a first integrated tracking system, including,
 - (i) a visual spectrum light source,
 - (ii) an infrared light source,
 - (iii) a camera;
- (b) mounting said first integrated tracking system on said mask, proximate a first eye of said diver so that said visual spectrum light source, said infrared light source, and said camera are pointing toward said first eye;
- (c) providing an electronics enclosure mounted on said mask, said electronics enclosure including a hub;
- (d) providing a remotely-located monitoring computer;
- (e) said integrated tracking system being connected to said hub;
- (f) said remotely-located monitoring computer being connected to said hub;
- (g) using said infrared light source and said camera to track movements of said first eye;

- (h) using said visual light source to provide a visual stimulus to said first eye while using said camera to monitor a response of said first eye to said visual stimulus; and
 - (i) using said remotely-located monitoring computer to monitor said first eye tracking functions and said visual stimulus and response.
2. The method for monitoring as recited in claim 1, wherein said connection between said integrated tracking system and said hub is made by a first cable and said connection between said remotely-located monitoring computer and said hub is made by a second cable.
3. The method for monitoring as recited in claim 1, further comprising:
- (a) providing a second integrated tracking system, including,
 - (i) a second visual spectrum light source,
 - (ii) a second infrared light source,
 - (iii) a second camera;
 - (b) mounting said second integrated tracking system on said mask, proximate a second eye of said diver so that said visual spectrum light source, said infrared light source, and said camera are pointing toward said second eye;
 - (c) said second integrated tracking system being connected to said hub;

- (d) using said second infrared light source and said second camera to track movements of said second eye; and
 - (e) using said second visual light source to provide a visual stimulus to said second eye while using said second camera to monitor a response of said second eye to said visual stimulus.
4. The method for monitoring as recited in claim 1, further comprising:
- (a) providing a second integrated tracking system, including,
 - (i) a second visual spectrum light source,
 - (ii) a second infrared light source,
 - (iii) a second camera;
 - (b) mounting said second integrated tracking system on said mask, proximate a second eye of said diver so that said visual spectrum light source, said infrared light source, and said camera are pointing toward said second eye;
 - (c) said second integrated tracking system being connected to said hub;
 - (d) using said visual spectrum light source to apply light to said first eye; and
 - (e) using said second integrated tracking system to monitor for a pupillary reflex in said second eye.

* * * * *